Exercises: Spectral Theory, SVD, and Matrix Decompositions

Mathematical Foundations for Machine Learning

Eigenvalues, Eigenvectors, and Diagonalization

Exercise 1.1 [Easy]

Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$.

A is diagonalizable PDP1 if it has n independent eigenvectors (forming P)

a) Find all eigenvalues and eigenvectors of A.

b) Is A diagonalizable? If so, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

c) Using the diagonalization, compute A^{10} (leave your answer in terms of P, D^{10}, P^{-1}, but compute D^{10} explicitly).

Exercise 1.2 [Easy]

True or False (with proof or counterexample):

a) If $A \in \mathbb{R}^{n\times n}$ has n distinct eigenvalues, then A is diagonalizable.

b) If A is diagonalizable, then A must have n distinct eigenvalues.

c) If $A^2 = A$, then the only possible eigenvalues of A are 0 and 1.

Exercise 1.3 [Medium]

Let $u, v \in \mathbb{R}^n$ be orthogonal unit vectors (i.e., $\|u\| = \|v\| = 1$ and $\langle u, v\rangle = 0$). Define

$$A = 3uu^T + 5vv^T.$$

!= 0 lives in span A. If = 0 then it lives in ker(A). If ker(A) = 0 then no 0, then ivertible

a) Prove that A is symmetric.

b) Find all eigenvalues and their corresponding eigenspaces.

c) What is $\operatorname{rank}(A)$? rank = non zero eigenvalues

d) Compute $\operatorname{tr}(A)$, $\det(A)$, and all singular values of A.

with eigenvalues

Exercise 1.4 [Medium] $1 \le \text{geometric} \le \text{algebraic}$ if geometric < algebraic then we don't hav

Let $A \in \mathbb{R}^{n\times n}$ and suppose A is diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_n$.

a) Prove that A^2 has eigenvalues $\lambda_1^2, \dots, \lambda_n^2$.

b) More generally, show that for any polynomial $p(t) = \sum_{k=0}^{m} c_k t^k$, the matrix $p(A)$ has eigenvalues $p(\lambda_1), \dots, p(\lambda_n)$.

c) Suppose A is invertible. Prove that A^{-1} has eigenvalues $\lambda_1^{-1}, \dots, \lambda_n^{-1}$.

Exercise 1.5 [Hard] diagonalizable if n indep. eigenvectors: PDP^-1. Eigenv if » has

Let $A \in \mathbb{R}^{n\times n}$ be a matrix such that $A^2 = A$ (i.e., A is idempotent).

a) Prove that A is diagonalizable. *Hi*

Sum $\ker(A) + \operatorname{im}(A)$: the set of all vectors $\mathbf{a} + \mathbf{b}$ where $\mathbf{a} \in \ker(A)$ and $\mathbf{b} \in \operatorname{im}(A)$. This is always a subspace.

b) Prove that $\operatorname{tr}(A) = \operatorname{rank}(A)$.

c) Now suppose additionally that A is symmetric. Prove that A is an orthogonal projection, i.e., $A = A^T = A^2$, and that all eigenvalues are exactly 0 or 1.

$V \oplus W$ is same as V+W but with V and W = {0}

3 properties for orthogonal projection

P: »=0 or »=1, then Av = 0 -> ker(A). Img(A) = v: v = Aw = A^2w = A(Aw) = Av

Symmetric Matrices and Positive (Semi-)Definiteness

Exercise 2.1 [Easy]

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$.

a) Verify that A is symmetric.
b) Find the eigenvalues of A. Is A positive definite, positive semi-definite, or neither?
c) Find an orthogonal matrix Q such that $A = Q\Lambda Q^T$.

Exercise 2.2 [Easy]

True or False (with proof or counterexample):

a) If every entry of $A \in \mathbb{R}^{n\times n}$ is positive, then A is positive definite.
b) If A is positive semi-definite, then every diagonal entry $a_{ii} \geq 0$.
c) If A and B are both positive semi-definite, then $A + B$ is positive semi-definite.

Exercise 2.3 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric and positive definite, and let $B \in \mathbb{R}^{n\times m}$ with $\text{rank}(B) = m$ (i.e., B has full column rank, with $m \leq n$).

a) Prove that $B^T AB \in \mathbb{R}^{m\times m}$ is symmetric.
b) Prove that $B^T AB$ is positive definite. take x^t -> y^tA^y -> all positive, B full rank then Bx != 0 then PD
c) Conclude that $B^T AB$ is invertible.

Exercise 2.4 [Medium]symmetric matrixes: geometric = algebraic -> always diagonalizable

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1, \ldots, \lambda_n$.

a) Show that A^2 is positive semi-definite.
b) Show that if A is positive definite, then A^{-1} is also positive definite.
c) Show that if A is positive semi-definite, then $\text{tr}(A) \geq 0$, and equality holds if and only if $A = 0$.

Exercise 2.5 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric and positive semi-definite. Define the map $\langle \cdot, \cdot \rangle_A : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

to be definitness you need all > 0 -> not inner pr.

$$\langle x, y \rangle_A = x^T Ay.$$

If matrix symmetric then by spectral theorem it's eigenspace spans all R^n -> so if A PSD + symmetric then x^tAx >= 0 for all x

-symmetry
-bilinearity
-non-negativity
-definitness

a) Show that $\langle \cdot, \cdot \rangle_A$ satisfies symmetry, bilinearity, and non-negativity.
b) Under what additional condition on A is $\langle \cdot, \cdot \rangle_A$ an inner product? Prove your claim.
c) Now assume $A \succ 0$. Prove the generalized Cauchy-Schwarz inequality:

$$|x^T Ay|^2 \leq (x^T Ax)(y^T Ay)$$

for all $x, y \in \mathbb{R}^n$. *Hint: Since $A \succ 0$, there exists a unique $A^{1/2} \succ 0$ with $A = A^{1/2}A^{1/2}$.*

Spectral Theorem and Variational Characterization

Exercise 3.1 [Easy]

Let $A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$.

Ra(x) takes any vector, not only eigenvectors -> between lambda 1 and lambda2

a) Find the eigenvalues $\lambda_1 \leq \lambda_2$ of A.
b) Compute the Rayleigh quotient $R_A(x) = \frac{x^T Ax}{x^T x}$ for $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Verify that $\lambda_1 \leq R_A(x) \leq \lambda_2$.
c) What is $\max_{\|x\|=1} x^T Ax$ and which x achieves it?

if you try to maximize with lagragian it gives you the eigen equaition

Exercise 3.2 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \le \lambda_2 \le \ldots \le \lambda_n$ and corresponding orthonormal eigenvectors $q_1, \ldots, q_n$.

a) Using the spectral decomposition $A = \sum_{i=1}^{n} \lambda_i q_i q_i^T$, show that

$$\max_{\|x\|=1} x^T A x = \lambda_n.$$

Hint: Write $x = \sum_{i=1}^{n} \alpha_i q_i$ with $\sum \alpha_i^2 = 1$.

b) Now consider the constrained problem:

$$\max_{\|x\|=1,\ x\perp q_n} x^T A x.$$

What is the maximum value? Prove your answer.

Exercise 3.3 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with spectral decomposition $A = Q\Lambda Q^T$. Define the matrix exponential $e^A = Q\,\mathrm{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})Q^T$.

a) Show that e^A is symmetric and positive definite.

b) Show that $\det(e^A) = e^{\mathrm{tr}(A)}$.

c) Show that $e^{A+B} = e^A e^B$ if A and B commute (i.e., $AB = BA$). *Hint: If $AB = BA$ and both are symmetric, they share an eigenbasis.*

Exercise 3.4 [Hard]

when you take the max , the x you choose is eigenvector n (qn). if you take x orth.to qn then

Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite with eigenvalues $0 < \lambda_1 \le \ldots \le \lambda_n$.

$$\lambda_k = \min_{\substack{V\subseteq\mathbb{R}^n \\ \dim(V)=n-k+1}} \max_{\substack{x\in V \\ \|x\|=1}} x^T A x$$

a) Prove using the variational characterization that

$$\min_{\|x\|=1} \|Ax\|_2 = \lambda_1, \qquad \max_{\|x\|=1} \|Ax\|_2 = \lambda_n.$$

equivalent to min max -> min over all possible subspaces dim that maximize

Hint: Consider $\|Ax\|_2^2 = x^T A^2 x$ and use the eigenvalues of A^2.

b) The condition number of A is defined as $\kappa(A) = \frac{\lambda_n}{\lambda_1}$. Show that $\kappa(A) = \|A\|_2 \|A^{-1}\|_2$, where $\|A\|_2 = \max_{\|x\|=1} \|Ax\|_2$.

$$\|A\|_2 = \max_{\|x\|=1} \|Ax\|_2$$

c) Solve the optimization problem

$$\max_{x\in\mathbb{R}^n,\ \|x\|_2=1} \|Ax\|_2 - \langle x, Ax\rangle.$$

Express the solution in terms of the eigenvalues of A.

Singular Value Decomposition and Low-Rank Perturbations

Exercise 4.1 [Easy]

Let $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \\ 0 & 0 \end{pmatrix}$.

a) Compute $A^T A$ and AA^T.

b) Find the singular values of A.

c) Write down the full SVD $A = U\Sigma V^T$.

Exercise 4.2 [Easy]

Let $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$ be nonzero vectors. Consider the rank-one matrix $A = uv^T$.

a) What is $\mathrm{rank}(A)$?

b) Find all singular values of A.
c) What are $\|A\|_2$ and $\|A\|_F$?

Exercise 4.3 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1, \ldots, \lambda_n$ (not necessarily non-negative).

a) Explain the relationship between the SVD of A and its spectral decomposition. How do the singular values relate to the eigenvalues?

b) Show that $\|A\|_F^2 = \sum_{i=1}^n \lambda_i^2$.

c) Let A_k denote the best rank-k approximation to A (in Frobenius norm). Express $\|A - A_k\|_F^2$ in terms of the eigenvalues of A.

Exercise 4.4 [Medium]

Let $A \in \mathbb{R}^{m\times n}$ have rank r with singular values $\sigma_1 \geq \ldots \geq \sigma_r > 0$. Let $E \in \mathbb{R}^{m\times n}$ be a perturbation with $\text{rank}(E) = p$.

a) Using the low-rank perturbation bound, we know that

$$\sigma_{k+p}(A) \leq \sigma_k(A - E) \quad \text{for } k = 1, \ldots, r - p.$$

Explain in words what this means.

b) Suppose A has rank $r = 5$ with singular values $\sigma_1 = 10, \sigma_2 = 8, \sigma_3 = 5, \sigma_4 = 3, \sigma_5 = 1$ and we subtract a rank-2 matrix E. What can you conclude about the singular values of $A - E$?

c) Conversely, show that for the best rank-k approximation A_k, we have $\|A - A_k\|_2 = \sigma_{k+1}(A)$.

Exercise 4.5 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric PSD with eigenvalues $\lambda_1 \geq \ldots \geq \lambda_n \geq 0$ and orthonormal eigenvectors $q_1, \ldots, q_n$. Consider the rank-one update $B = A + \gamma q_1 q_1^T$ for some $\gamma > 0$.

a) Find all eigenvalues and eigenvectors of B.

b) How do the singular values of B relate to those of A?

c) Now consider a general rank-one update $C = A + \gamma w w^T$ where $w \in \mathbb{R}^n$ is a unit vector not aligned with any eigenvector of A. Without computing explicitly, explain why the eigenvalues of C interlace with those of A:

$$\mu_1 \geq \lambda_1 \geq \mu_2 \geq \lambda_2 \geq \ldots \geq \mu_n \geq \lambda_n$$

where μ_i are the eigenvalues of C in decreasing order.

Power Method

Exercise 5.1 [Easy]

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and starting vector $x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

a) Perform 2 iterations of the power method: compute $x_1 = \frac{Ax_0}{\|Ax_0\|}$ and $x_2 = \frac{Ax_1}{\|Ax_1\|}$.

b) Estimate the dominant eigenvalue using the Rayleigh quotient $R_A(x_k)$ at each step.

c) What are the true eigenvalues? To which eigenvector does the power method converge?

Exercise 5.2 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $|\lambda_1| > |\lambda_2| \geq \ldots \geq |\lambda_n|$ and orthonormal eigenvectors $q_1, \ldots, q_n$.

a) Write the starting vector $x_0 = \sum_{i=1}^{n} \alpha_i q_i$ with $\alpha_1 \neq 0$. Show that after k multiplications (without normalization):

$$A^k x_0 = \lambda_1^k \left(\alpha_1 q_1 + \sum_{i=2}^{n} \alpha_i \left(\frac{\lambda_i}{\lambda_1} \right)^k q_i \right).$$

b) Explain why $\frac{A^k x_0}{\|A^k x_0\|} \to \pm q_1$ as $k \to \infty$.

c) What is the rate of convergence? What happens if $|\lambda_1| = |\lambda_2|$?

Exercise 5.3 [Hard]

Let $A \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues $\lambda_1 > \lambda_2 > \ldots > \lambda_n$. Suppose we already know λ_1 and q_1 (from the power method). We want to find λ_2 and q_2.

a) Define $B = A - \lambda_1 q_1 q_1^T$ (deflation). Show that B has eigenvalues $0, \lambda_2, \ldots, \lambda_n$ with the same eigenvectors as A.

b) Why can we not always directly apply the power method to B to find λ_2? What condition on λ_2 relative to the other eigenvalues is needed?

c) Propose a modification: use the *inverse power method* with shift μ close to λ_2, i.e., iterate $y_{k+1} = (B - \mu I)^{-1} y_k$ (normalized). Explain why this converges to the eigenvector for λ_2 and what determines the convergence rate.

Pseudo-Inverse, Cholesky, Generalized Eigenproblems, Von Neumann

Exercise 6.1 [Easy]

Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ (full column rank).

a) Compute the pseudo-inverse $A^\dagger$ using the formula $A^\dagger = (A^T A)^{-1} A^T$.

b) Verify that $A^\dagger A = I_2$.

c) What is $AA^\dagger$? Is it an orthogonal projection? Onto what subspace?

Exercise 6.2 [Medium]

Let $A = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$.

a) Verify that A is positive definite.

b) Find the Cholesky factorization $A = LL^T$ where L is lower triangular with positive diagonal entries.

c) Use the Cholesky factorization to solve $Ax = b$ where $b = \begin{pmatrix} 8 \\ 11 \end{pmatrix}$. (First solve $Ly = b$, then $L^T x = y$.)

Exercise 6.3 [Medium]

Let $A, B \in \mathbb{R}^{n \times n}$ with B symmetric positive definite. Consider the **generalized eigenvalue problem** $Ax = \lambda Bx$.

a) Show that this can be reduced to a standard eigenvalue problem by substituting $z = B^{1/2} x$. What is the resulting standard eigenproblem?

b) Let $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. Find the generalized eigenvalues.

c) Show that the generalized Rayleigh quotient $R_{A,B}(x) = \frac{x^T A x}{x^T B x}$ achieves its maximum at the largest generalized eigenvalue.

Exercise 6.4 [Medium]

(Von Neumann Trace Inequality) Let $A, B \in \mathbb{R}^{n\times n}$ with singular values $\sigma_1(A) \geq \ldots \geq \sigma_n(A)$ and $\sigma_1(B) \geq \ldots \geq \sigma_n(B)$ respectively. The von Neumann trace inequality states:

$$|\operatorname{tr}(A^T B)| \leq \sum_{i=1}^{n} \sigma_i(A)\sigma_i(B).$$

a) Verify this inequality for $A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$.

b) Show that equality holds when A and B share the same left and right singular vectors, i.e., $A = U\Sigma_A V^T$ and $B = U\Sigma_B V^T$.

c) Use the von Neumann trace inequality to show that for any $A \in \mathbb{R}^{n\times n}$: $|\operatorname{tr}(A)| \leq \sum_{i=1}^{n} \sigma_i(A)$.

Mixed Challenging Exercises

Exercise 7.1 [Hard]

Let $u, v \in \mathbb{R}^n$ with $\|u\| = 2$, $\|v\| = 1$, and $\langle u, v\rangle = 1$. Define

$$A = uu^T - vv^T.$$

a) Prove that A is symmetric. What is $\operatorname{rank}(A)$?

b) The matrix A acts nontrivially only on $\operatorname{span}(u, v)$. Find the 2×2 matrix representing A restricted to $\operatorname{span}(u, v)$ with respect to the basis $\{u, v\}$, and find all eigenvalues of A.

c) Compute $\operatorname{tr}(A)$, $\det(A)$, and all singular values of A.

d) Solve $\max\limits_{x\in\mathbb{R}^n,\ \|x\|_2=1} \|Ax\|_2$.

Exercise 7.2 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite.

a) Show that A has a unique Cholesky decomposition $A = LL^T$ where L is lower triangular with positive diagonal entries. *Hint: Use induction on n with the block structure* $A = \begin{pmatrix} a_{11} & w^T \\ w & A_{22} \end{pmatrix}$.

b) Let A have eigenvalues $\lambda_1 \geq \ldots \geq \lambda_n > 0$. Express the condition number $\kappa(A)$ in terms of the singular values of the Cholesky factor L.

c) Show that the pseudo-inverse of A equals A^{-1} when A is invertible, by verifying all four Moore-Penrose conditions.

Exercise 7.3 [Hard]

Let $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ and $\operatorname{rank}(A) = n$ (full column rank). Consider the least squares problem $\min_x \|Ax - b\|_2^2$.

a) Show that the solution is $x^* = A^\dagger b = (A^T A)^{-1} A^T b$.

b) Show that $P = AA^\dagger = A(A^T A)^{-1} A^T$ is an orthogonal projector onto $\operatorname{im}(A)$. Verify $P^2 = P$ and $P^T = P$.

c) Now suppose $A = U\Sigma V^T$ is the thin SVD (with $U \in \mathbb{R}^{m\times n}$, $\Sigma \in \mathbb{R}^{n\times n}$, $V \in \mathbb{R}^{n\times n}$). Express x^* and P in terms of U, Σ, V.

d) Show that the residual $r = b - Ax^*$ satisfies $r \perp \operatorname{im}(A)$.

Exercise 7.4 [Hard]

Let $A \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues $\lambda_1 \leq \ldots \leq \lambda_n$ and let $E = \gamma w w^T$ where $w \in \mathbb{R}^n$ is a unit vector and $\gamma > 0$.

a) What is $\text{rank}(E)$? What is $\|E\|_2$?

b) Let $\mu_1 \leq \ldots \leq \mu_n$ be the eigenvalues of $A+E$. Using the interlacing property for rank-one perturbations, give the tightest bounds you can on each μ_i.

c) Now suppose A has a spectral gap: $\lambda_n - \lambda_{n-1} = \Delta > 0$, and $\gamma < \Delta$. Show that μ_n remains the unique largest eigenvalue of $A + E$, and give a bound on $|\mu_n - \lambda_n|$.

d) Explain why this result is relevant for the convergence of the power method applied to $A + E$ versus A.